

Reproducibility Test

Week – 4

DS-1

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1. Datasets:

All the previous used datasets from Week 1 to 3 are loaded into a dataset folder.

2. Scripts:

All the previous tasks notebooks converted into scripts and these scripts are used to save results into Outputs folder.

3. Master runner:

New file “**run_all.py**” is created and after this file run the all the scripts gets executed and results are stored in outputs folder.

4. Outputs:

All the results of tasks 1-3 of all weeks stored in outputs folder.

5. Log file:

Reproducibility test run by run_all.py file and all the files from Week 1 to Week 3 are executed and a log file is stored with execution time.

NHS Mental Health DS Pipeline - Full Run

```
--> Week 1 DS-1 (scripts/week1/ds1_statistical_baseline_model.py)
Done in 13.4s -> APMS2014_regional_prevalence.png

--> Week 1 DS-2 (scripts/week1/ds2_hypothesis_testing.py)
Done in 2.0s -> t-test Male vs Female CMD rates Equal CMD rates by sex -2.8099 0.0158 Re

--> Week 1 DS-3 (scripts/week1/ds3_distribution_analysis.py)
Done in 5.7s -> dist_regional_cmd.png | dist_temporal_trend.png

--> Week 2 DS-1 (scripts/week2/ds1_regression_modelling.py)
Done in 4.7s -> -> regression_metrics.csv | regression_coefficients.csv | regression_results.png

--> Week 2 DS-2 (scripts/week2/ds2_clustering_patient_segments.py)
Done in 10.6s -> -> clustered_patients.csv | cluster_profiles.csv | cluster_analysis.png

--> Week 2 DS-3 (scripts/week2/ds3_temporal_pattern_detection.py)
Done in 4.3s -> -> temporal_rolling_averages.png | temporal_chow_scan.png | temporal_chow_test_res

--> Week 3 DS-1 (scripts/week3/ds1_iapt_demand_forecasting.py)
Done in 5.9s -> -> forecast_model_comparison.png | forecast_3month_ahead.png | forecast_metrics.cs

--> Week 3 DS-2 (scripts/week3/ds2_prescribing_anomaly_detection.py)
Done in 5.2s -> -> anomaly_flags.csv | anomaly_overview.png | anomaly_heatmap.png

--> Week 3 DS-3 (scripts/week3/ds3_geo_spatial_analysis.py)
Done in 5.1s -> -> spatial_imd_map.png | spatial_access_rate_map.png | spatial_underserved_map.png
```

Summary

Week 1 DS-1	OK	13.4s
Week 1 DS-2	OK	2.0s
Week 1 DS-3	OK	5.7s
Week 2 DS-1	OK	4.7s
Week 2 DS-2	OK	10.6s
Week 2 DS-3	OK	4.3s
Week 3 DS-1	OK	5.9s
Week 3 DS-2	OK	5.2s
Week 3 DS-3	OK	5.1s

9/9 scripts passed | Total time: 57.0s

All outputs saved to outputs/